

Panagiotis C. Theocharopoulos

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Current Positions

- 2025 – Present 📌 **Lead AI Engineer**, Covariance, Athens, Greece
- 2024 – Present 📌 **Research Fellow**, University of Thessaly, Greece
- 2021 – Present 📌 **PhD candidate** at University of Thessaly, Greece

Professional Experience

01/2025 – Present 📌 **Lead AI Engineer**, Covariance, Athens, Greece

- Leading and managing a high-performing data science team focused on healthcare analytics and AI solutions.
- Overseeing the development of LLM and RAG systems, directing multiple ML projects, and driving technical innovation.
- Responsible for defining technical standards, mentoring team members, and fostering a collaborative environment to achieve project milestones and organizational goals.

02/2024 – 01/2025 📌 **Senior Data Scientist**, Covariance, Athens, Greece

- Led end-to-end machine learning projects and architected advanced LLM and RAG solutions for cutting-edge AI applications.
- Played a key role in securing funding by authoring proposals for national and European programs.
- Enhanced team processes by implementing technical best practices and delivering impactful AI-driven insights in healthcare.

02/2021 – 02/2024 📌 **Data Scientist**, Covariance, Athens, Greece

- Executed data gathering, ETL processes, data wrangling, and statistical analysis.
- Built and deployed machine learning models, NLP systems, and web-based data science applications.
- Collaborated across teams to deliver scalable AI solutions tailored to business needs.

2019 – 2023 📌 **Website Administrator**, University of Thessaly, Department of Mathematics, Lamia, Greece

- Led website development and design, managed databases and servers, and provided technical support for university digital infrastructure.
- Ensured uptime and security for all online assets.

Professional Experience (continued)

07/2017 – 08/2017  **Information Systems Support Engineer (Internship)**, Naval Hospital of Athens, Athens, Greece

- Maintained and optimized servers (Windows, AIX), provided network support for WAN & LAN (up to 2000 users), and managed MySQL databases.
- Developed and sustained local web applications, configured network devices (CISCO, HP, Mikrotik), and implemented robust security policies (ACL, Firewall, NAT, VPN).
- Ensured smooth operation of the Naval Telemedicine System with a focus on connectivity and security.

Education

- 2021 – Present  **PhD Candidate**, Department of Computer Science and Biomedical Informatics, University of Thessaly, Greece
Thesis title: *“Development of Deep Learning methods on big data with a focus on Natural Language Processing”*
- 2020  **M.Sc.** Informatics and Computational Biomedicine, University of Thessaly, Greece
Minor: Computational Medicine and Biology
Master thesis: *“The EU’s General Data Protection Regulation and the Data Protection Officer in Biomedical Applications”*
- 2019  **Ptychion (4-year Bachelor)** Department of Informatics and Computational Biomedicine, University of Thessaly, Greece
- 2017  **Erasmus+ Exchange Student** Università degli Studi di Perugia, Perugia, Italy

Scholarships / Awards / Awarded Funding

- 2025  **IEEE CIS Student Travel Grant Award**
For the IEEE Symposium Series on Computational Intelligence 2025, (SSCI 2025), Trondheim, Norway, 2025
- 2022  **Best National Startup of 2022 (Elevate Greece)**
Awarded by Elevate Greece to Covariance.¹
- 2019 – 2021  **Academic Work Scholarship**
University of Thessaly
- 2019  **Tuition Fee Exemption Scholarship**
Awarded for M.Sc. degree.
- 2016 – 2017  **Erasmus+ Exchange program**
Participated in the Erasmus+ program.

¹ As an employee of Covariance.

Participation in Research Projects

- 2024 – 2025  **Bridging large omics, genetic, and medical data for the wide application of Precision Medicine in Greece** (TAEDR–0539180)
Role: Research Fellow
Institutions: University of Thessaly, B.S.R.C. Fleming, University of Patras, Hellenic Pasteur Institute
Funding Source: National Recovery and Resilience Plan (Greece 2.0), Recovery and Resilience Fund
Funding Agency: European Union – NextGenerationEU

Teaching Experience

- 2023 – 2025  Teaching assistant for **postgraduate** courses, University of Thessaly, Greece:
- Autumn semester: **Data Mining and Big Data Analysis in Medicine and Biology**
 - Spring semester: **Programming Subjects**
- 2019 – 2024  Teaching assistant for **undergraduate** courses, Department of Mathematics, University of Thessaly, Greece:
- Autumn semester: **Introduction to Programming I**
 - Spring semester: **Introduction to Programming II**
 - Spring semester: **Numerical Analysis**
- 2022 – 2023  Teaching assistant for **undergraduate** course, Department of Computer Science and Biomedical Informatics, University of Thessaly, Greece:
- Autumn semester: **Databases**

Reviewer

-  **Neural Computing and Applications** (Springer)
-  **International Journal of Information Technology & Decision Making** (World Scientific)
-  **Public Health** (ScienceDirect)
-  **Frontiers in Public Health**
-  **2025 IEEE Symposium Series on Computational Intelligence (SSCI)**
-  **2025 International Joint Conference on Neural Networks (IJCNN)**
-  **26th Engineering Applications and Advances of Artificial Intelligence (EAAAI 2025)**
-  **22nd IEEE Conference on CI/AI in Bioinformatics and Computational Biology (CIBCB 2025)**
-  **20th International Conference on Availability, Reliability, and Security (ARES 2025)**
-  **Pacific Symposium on Biocomputing (PSB) 2026**

Publications

Refereed Journal Publications

- [J1] **P. C. Theocharopoulos**, P. Anagnostou, S. V. Georgakopoulos, S. K. Tasoulis, and V. P. Plagianakos, “Large language models for efficient topic modeling,” *Neural Computing and Applications*, Sept. 2025, doi: 10.1007/s00521-025-11593-9.
- [J1] **P. C. Theocharopoulos**, S. Bersimis, S. V. Georgakopoulos, A. Karaminas, S. K. Tasoulis, and V. P. Plagianakos, “Developing predictive precision medicine models by exploiting real-world data using machine learning methods” *Journal of Applied Statistics*, 2024, doi: 10.1080/02664763.2024.2315451.
- [J2] **P. C. Theocharopoulos**, A. Tsoukala, S. V. Georgakopoulos, S. K. Tasoulis, and V. P. Plagianakos, “Analysing sentiment change detection of Covid-19 tweets”, *Neural Computing and Applications*, pp. 1–11, 2023.

Refereed Conference Publications

- [C1] **P. C. Theocharopoulos**, S. Bersimis, S. V. Georgakopoulos, V. P. Plagianakos and S. K. Tasoulis, “Transforming Medical Practice: Harnessing the Power of Big Data and Machine Learning for Predictive Precision Medicine,” 2025 IEEE Symposium on Computational Intelligence in Health and Medicine (CIHM), Trondheim, Norway, 2025, pp. 1–7, doi: 10.1109/CIHM64979.2025.10969489.
- [C2] **P. C. Theocharopoulos**, S. V. Georgakopoulos, S. K. Tasoulis, and V. P. Plagianakos, “Who Writes the Review, Human or AI?” , in *IEEE 22nd International Conference on Cognitive Informatics and Cognitive Computing (ICCI-CC’23)*, Stanford, CA, USA, 2023. (Proceedings not published)
- [C3] **P. C. Theocharopoulos**, P. Anagnostou, A. Tsoukala, S. V. Georgakopoulos, S. K. Tasoulis, and V. P. Plagianakos, “Detection of Fake Generated Scientific Abstracts”, in *2023 IEEE Ninth International Conference on Big Data Computing Service and Applications (BigDataService)*, Athens, Greece, 2023.
- [C4] **P. C. Theocharopoulos**, A. Tsoukala, S. V. Georgakopoulos, S. K. Tasoulis, and V. P. Plagianakos, “Text analysis of COVID-19 tweets”, in *International Conference on Engineering Applications of Neural Networks*, Springer, 2022, pp. 517–528.

Skills

Languages	■ Greek, English, German, Italian
Coding	■ Python, R, SQL, L ^A T _E X, MATLAB, Power BI, WordPress, ...
Operating Systems	■ Linux-based and Unix-based OS, Microsoft Windows

Certification

- 2024 ■ **Building RAG Agents with LLMs**
Issued by NVIDIA
- **I2I Spin Out Innovation Programme**
Issued by the University of Galway

Certification (continued)

- **Power BI Financial Reporting & Financial Analysis: A to Z**
Issued by Udemy
- 2022 ■ **EHDEN's OMOP-CDM Data Standardization and ETL Processes²**
Issued by European Health Data & Evidence Network (EHDEN)
- 2019 ■ **Pedagogical and Teaching Competence**
Issued by the University of Thessaly

²As an employee of Covariance